Problem Set 5 – Cost Drivers in Childcare

MGT-403 Data Project · Where Should We Open Our Next Daycare?

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Summary of answers

- **Q1–Q3 data.** We built a year × county panel of 1,233–1,235 metro counties per year (2015–2022) by combining ACS 5-year county data from the Census API with BLS OEWS childcare-worker (SOC 39-9011) area wages through the course crosswalk. Wages are missing for 0.4–1.6% of metro counties per year, in line with the roughly 1% the assignment says to expect.
- **Q4 (2022).** The average metro county has a childcare-worker mean annual wage of $27,373 (SD $4,337) and a median gross rent of $1,052/month (SD $342).
- **Q5 (real trends).** In constant 2022 dollars, the average childcare-worker wage rose from $25,851 (2015) to a peak of $27,646 (2021), then slipped to $27,373 in 2022 (+5.9% over 2015–2022). Real rent rose every year, from $993 to $1,052 per month (+5.9%).
- **Q6 (2022 vs 2019).** Real wages were $456.36 higher in 2022 (+1.7%). Paired t = 8.58, p = 2.93×10^{-17}, so we reject H_0 that real wages were the same. The rise is statistically significant but small in size: wages beat inflation by under 2%.
- **Q7 (regression).** Wage = $22,359 + 16,500 × female college share (t = 15.6, p = 4.48×10^{-50}, $R^2 = 0.168$). We reject that the slope is zero: counties where 10 percentage points more women hold a bachelor's degree pay childcare workers about $1,650 more per year, on average.

Q1. ACS 5-year county data (Census API)

We pulled the ACS 5-year estimates for every U.S. county for each vintage from 2015 to 2022 from `api.census.gov/data/{year}/acs/acs5` (`for=county:*&in=state:*`), using the variable codes in the assignment's table. The year is the end year of the 5-year window (e.g., 2022 = 2018–2022 responses). Before using them, we downloaded the Census's own description of every code for all eight vintages. They are identical across years, apart from the dollar-year in the income label. County FIPS = state (2 digits) + county (3 digits), kept as text. Census “not available” codes (large negative numbers) are set to missing.

Derived variable	Construction (ACS codes)	Census label (2022)
per_capita_income	B19301_001E	Per capita income in the past 12 months (in 2022 inflation-adjusted dollars)
pop_total	B01001_001E	Total
kids_under6	B23008_002E	Under 6 years
kids_under6_allpar_lf	B23008_004E + 010E + 013E	Under 6 › two parents, both in LF; + one parent (father) in LF; + one parent (mother) in LF
median_gross_rent	B25064_001E	Median gross rent
female_college_share	(B15002_032E+033E+034E+035E) / B15002_019E	Female 25+ with bachelor's, master's, professional, or doctorate ÷ all females 25+
female_ftyr_share	B23022_029E / B23022_026E	Females 16–64 who usually worked 35+ hrs/wk, 50–52 weeks ÷ all females 16–64
under6_share	kids_under6 ÷ pop_total	as defined in the assignment
under6_careneed_share	kids_under6_allpar_lf ÷ pop_total	as defined in the assignment

Table 1. ACS variables. Counties returned per vintage: 2015: 3,220, 2016: 3,220, 2017: 3,220, 2018: 3,220, 2019: 3,220, 2020: 3,221, 2021: 3,221, 2022: 3,222.

Q2. OEWS childcare-worker wages by area

From each `oesmYYma.zip` (May 2015–May 2022) we kept the rows for SOC **39-9011 "Childcare Workers"** and the **mean annual wage (A_MEAN)**, plus the mean hourly wage. Each zip contains a metropolitan file (*MSA_*) and a nonmetropolitan file (*BOS_*). **Important detail:** in 2015–2017, the 11 largest metros (New York, LA, Chicago, Boston, Washington, etc.) appear in the MSA_ file only split into metropolitan divisions. The whole-metro estimates that the crosswalk refers to are in a third file, *aMSA_*. Leaving out aMSA_ left about 10% of metro counties without a wage in 2015–2017, so we included it. No area appears twice in any year. Column names differ across years (e.g., AREA_NAME vs. area_title), so we standardized them. BLS marks suppressed wages with "*"; these are treated as missing.

Year	Childcare-worker rows by file (MSA_ / aMSA_ / BOS_)
2015	BOS (157), MSA (415), aMSA (11)
2016	BOS (157), MSA (415), aMSA (11)
2017	BOS (158), MSA (417), aMSA (11)
2018	BOS (133), MSA (392)
2019	BOS (132), MSA (392)
2020	BOS (133), MSA (384)
2021	BOS (135), MSA (389)
2022	BOS (135), MSA (384)

Table 2. SOC 39-9011 rows extracted from each OEWS release.

Q3. Merge to a year × county metro panel

Steps: (1) zero-pad the OEWS area code to 7 characters (e.g., 16980 → 0016980) so it matches the crosswalk; (2) keep crosswalk rows whose *oews_area* begins with "00" (metro); (3) join OEWS wages on [year, oews_area]; (4) join ACS on [county_fips, year]; (5) label metros with the crosswalk's *area_title*. The result has exactly one row per year × county (9,873 rows).

Data-handling decision. The crosswalk lists two retired county codes next to their current replacements: 12025 (old Dade County, FL code; now 12086 Miami-Dade) and 51515 (Bedford city, VA; now part of 51019 Bedford County). They never appear in any ACS vintage and would count their metro's wage twice, so we dropped them.

Year	Metro counties	No OEWS row	Wage missing	% wage missing	No ACS match	Rent missing
2015	1,233	10	11	0.89%	0	1
2016	1,233	13	14	1.14%	0	0
2017	1,234	10	10	0.81%	0	0
2018	1,233	5	5	0.41%	0	0
2019	1,235	6	6	0.49%	0	0
2020	1,235	18	20	1.62%	0	0
2021	1,235	13	13	1.05%	0	1
2022	1,235	20	20	1.62%	7	8

Table 3. Merge sanity check. "No OEWS row" = BLS published no SOC 39-9011 estimate for that metro and year; "Wage missing" also counts "*"-suppressed cells.

- **Wages:** 0.41–1.62% missing per year, matching the assignment's "roughly 1%". Missing cases are metros for which BLS published no childcare-worker estimate or suppressed it (2022: Arecibo, PR, Bremerton-Silverdale, WA, Columbus, IN, Fond du Lac, WI, Gadsden, AL, Guayama, PR, Homosassa Springs, FL, Janesville-Beloit, WI, Madera, CA, Mayaguez, PR, Pine Bluff, AR, Yuma, AZ).
- **ACS:** every metro county matches except **7 Connecticut counties in 2022**. Starting with the 2022 ACS, the Census Bureau reports Connecticut by 9 planning regions (FIPS 09110–09190) instead of its old counties, while the crosswalk still uses the old county codes. These 7 county-years are left missing for ACS variables (no county-to-region allocation was made).

Q4. Descriptive statistics of the predictors, 2022

Variable	Mean	SD	N	Min	Max
Childcare-worker mean annual wage, OEWS ($)	27,373	4,337	1,215	19,540	41,700
Childcare-worker mean hourly wage, OEWS ($)	13.16	2.08	1,215	9.39	20.05
Per-capita income ($)	36,156	10,504	1,228	8,318	89,702
Total population	231,863	506,835	1,228	50	9,936,690
Own children under 6	15,441	34,275	1,228	0	616,046
Under-6 w/ all resident parents in labor force	10,420	22,637	1,228	0	406,465
Median gross rent ($/month)	1,052	342	1,227	330	2,805
Share women 25+ with bachelor's+	0.3013	0.1070	1,228	0.0812	0.7761
Share women 16-64 full-time year-round	0.4367	0.0683	1,228	0.1648	0.7143
Share kids under 6 (of population)	0.0629	0.0121	1,228	0.0000	0.1093
Share kids under 6 who need care (of population)	0.0423	0.0094	1,228	0.0000	0.0695

Table 4. Metro counties, 2022 (OEWS May 2022; ACS 2018–2022). 1,235 metro counties; the wage comes from 377 OEWS metro areas and is shared by all counties in an area. Dollar values are 2022 dollars. Shares are fractions (0.30 = 30%).

- Childcare work is low-paid: the average metro county's childcare-worker mean wage is $27,373 per year ($13.16/hour), ranging from $19,540 to $41,700.
- Population and child counts are extremely skewed (SD far above the mean): metro counties range from tiny (Kalawao County, HI, population 50, part of the Kahului metro) to Los Angeles County (~9.9 million). The share variables are better for comparing counties.
- On average 30.1% of women 25+ have a bachelor's degree or more, 43.7% of women 16–64 work full-time year-round, and children under 6 are 6.3% of the population (4.2% are under-6s whose resident parents are all in the labor force).

Q5. Real wages and rents over time (constant 2022 dollars)

Real value$_t$ = nominal value$_t$ × CPI$_{2022}$ / CPI$_t$ (CPI-U, CUUR0000SA0, annual average). OEWS wages are treated as year-t dollars (May reference period). ACS dollar values are already expressed in dollars of the 5-year window's end year (the Census label says so, e.g., "in 2015 inflation-adjusted dollars"), so they are also treated as year-t dollars. Lines show the mean across metro counties, with a 95% CI. The dashed line is a balanced panel (counties observed in every year).

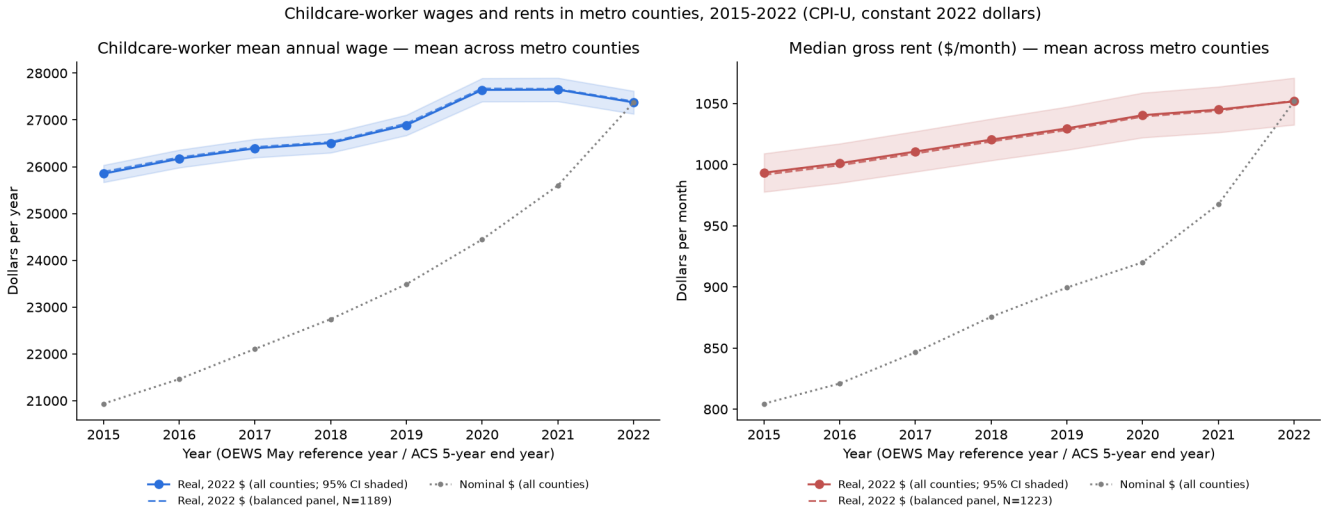


Figure 1. Childcare-worker mean annual wage (left) and median gross rent (right), metro counties.

Year	CPI-U	Wage nominal	Wage real (2022 $)	N	Rent nominal	Rent real (2022 $)	N
2015	237.017	$20,936	$25,851	1,222	$805	$993	1,232
2016	240.007	$21,462	$26,170	1,219	$821	$1,001	1,233
2017	245.120	$22,106	$26,393	1,224	$847	$1,011	1,234
2018	251.107	$22,743	$26,506	1,228	$876	$1,020	1,233
2019	255.657	$23,490	$26,889	1,229	$899	$1,030	1,235
2020	258.811	$24,445	$27,641	1,215	$920	$1,040	1,235
2021	270.970	$25,598	$27,646	1,222	$968	$1,045	1,234
2022	292.655	$27,373	$27,373	1,215	$1,052	$1,052	1,227

Table 5. Mean across metro counties; wage in $/year, rent in $/month.

Discussion

- **Wages:** nominal wages rose 30.7% from 2015 to 2022, while the CPI rose 23.5%. In real terms, wages rose gradually through 2019, jumped in 2020 (+2.8%), were flat in 2021, and then fell 1.0% in 2022 as 8.0% inflation outpaced nominal raises. Net change 2015–2022: +5.9% real.
- **Rents:** real median gross rent rose every year, +5.9% in total ($993 → $1,052/month). Because each ACS figure pools five years of responses, the rent series is smoothed and reacts slowly to short-run changes.
- The balanced-panel lines lie almost on top of the all-county lines, so these trends are not caused by changes in which counties are in the sample.

Q6. Were real childcare-worker wages the same in 2022 as in 2019?

(a) Hypotheses. Let μ_{2019}, μ_{2022} be the average real (2022 $) childcare-worker mean annual wage across metro counties.

$$H_0: \mu_{2022} - \mu_{2019} = 0 \quad H_1: \mu_{2022} - \mu_{2019} \neq 0 \quad \text{(two-sided, } \alpha = 0.05\text{)}$$

(b) Test. The same metro counties are observed in both years, so the main test is a **paired t-test** on each county's change d = real wage$_{2022}$ − real wage$_{2019}$ (2019 wages converted with CPI_{2022}/CPI_{2019} = 292.655/255.657). As checks, we also ran a two-sample Welch t-test on all counties in each year, and a paired test with one observation per OEWS metro area (the level at which BLS measures wages).

$$t = \bar{d} / (s_d / \sqrt{n}) = 456.3581 / (1852.3232 / \sqrt{1212}) = 456.3581 / 53.2066 = \mathbf{8.5771}, \quad df = 1211; \quad p = 2 \times P(T_{1211} \geq |t|) = \mathbf{2.93 \times 10^{-17}}$$

	Paired, counties (main)	Welch two-sample, counties	Paired, metro areas
n	1,212 counties in both years	1,229 (2019), 1,215 (2022)	374 metro areas
Mean real wage 2019	$26,912.70	$26,888.87	$27,207.57
Mean real wage 2022	$27,369.06	$27,373.48	$27,704.73
Difference (2022 − 2019)	$456.36	$484.61	$497.16
t statistic	8.577	2.904	4.457
p-value (two-sided)	2.93×10^{-17}	0.0037	1.10×10^{-5}
Decision at 5%	Reject H_0	Reject H_0	Reject H_0
Share of units with a real raise	62.2%	–	60.7%

Table 6. Real (2022 $) childcare-worker mean annual wage, 2022 vs 2019. Paired 95% CI for the change: [$351.97, $560.75]. SciPy's ttest_rel gives t = 8.5771, identical to the hand formula; a separate program reading the raw files reproduced n, mean change and t.

(c) Interpretation

- **Reject H_0.** |t| = 8.58 is far above the critical value of 1.96, and p = 2.93×10^{-17} < 0.05. Real childcare-worker wages were significantly higher in 2022 than in 2019, by $456 per year (+1.7%). Wages rose in 62% of counties. The two checks agree (Welch p = 0.0037; metro-area p = 1.10×10^{-5}).
- **The rise is statistically clear but economically small.** Nominal wages rose 16.4% (from $23,510 to $27,369), but prices rose 14.5%, so most of the nominal raise only kept pace with inflation. Real wages also peaked in 2020–2021 and fell in 2022 (Q5).
- **What this means for the providers' claim.** The data support part of it: childcare workers' pay rose faster than general inflation after 2019, and in dollar terms labor costs rose about 16%. But the real increase is under 2%, which is modest for something described as an “important driver”. For context, from our PS4 results: real median preschool prices in a fixed set of counties went from $174.01 (2019) to $168.99 (2022), -2.9%. So real labor costs rose a little while real prices fell. A comparison of averages cannot show that wages *caused* cost increases, and it says nothing about other costs such as rent (which also rose, Q5).

Q7. Bivariate regression (2022): wages on female college share

Model: $wage_c = \beta_0 + \beta_1 \cdot female_college_share_c + u_c$, estimated by OLS on 1,208 metro counties with both variables in 2022 (wage = OEWS mean annual wage, 2022 $; share is a fraction from 0 to 1).

	Coefficient	Std. error	t	p-value	95% CI
Intercept (β_0)	$22,359.19				
female_college_share (β_1)	16,499.99	1,058.19	15.59	4.48×10^{-50}	[14,424, 18,576]

	Coefficient	Std. error	t	p-value	95% CI
robust (HC1) SE		1,092.45	15.10	1.53×10^{-51}	
clustered by metro area (373 clusters)		2,217.33	7.44	9.96×10^{-14}	

Table 7. OLS, N = 1,208, R^2 = 0.168, correlation = 0.410. Hand check: slope = cov(x, y)/var(x) = 16,499.99.

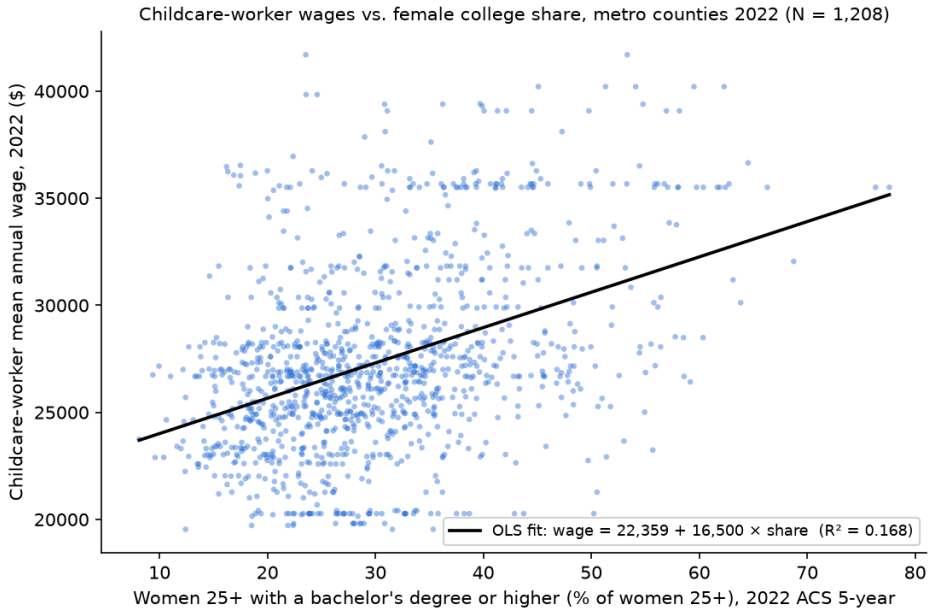


Figure 2. Each dot is a metro county. Horizontal bands appear because all counties in a metro area share one OEWS wage.

- **Coefficient:** β_1 = 16,500. Since the share runs from 0 to 1, this means a **10 percentage-point** higher share of college-educated women is associated with a **$1,650** higher childcare-worker mean annual wage (about 6% of the average wage). Moving 1 SD in the share (10.7 points) goes with $1,773 higher wages.
- **Can we reject $\beta_1 = 0$? Yes.** t = 15.6, p = 4.48×10^{-50}, and the 95% CI [14,424, 18,576] excludes zero. Counties in the same metro share one wage, so they are not independent observations. Allowing for that (clustered SEs) doubles the standard error, but t is still 7.4, so the conclusion holds.
- **Interpretation in words:** childcare workers earn more where more women are college-educated. This is a correlation, not a causal effect, and it explains about 17% of the variation in wages. The share of college-educated women likely also reflects higher-wage, higher-cost local labor markets (interpretation, not tested here).

Assumptions and data-handling choices

- Sample: crosswalk counties whose OEWS area code begins with “00” (metro), 2015–2022, the same metro definition as PS4. Each county counts equally (not population-weighted).
- Wage measure: OEWS mean annual wage (A_MEAN) for SOC 39-9011; the year is the May reference year. 2015–2017 whole-metro wages for the 11 largest metros come from the aMSA_ files.
- Two retired FIPS codes (12025, 51515) were dropped as duplicates of current counties. Connecticut's 7 metro counties lack ACS data in 2022 because the 2022 ACS uses planning regions.
- Dollar conversion: CPI-U annual averages from the BLS API (2015–2022), base year 2022.
- Q6 main test: paired t-test (same counties both years); Welch and metro-area versions as checks. Q7: conventional OLS SEs, with robust and metro-clustered SEs as checks.
- Reproduce: put the 8 OEWS zips in data/oews/, the crosswalk and CPI file in data/, and your Census key in data/census_api_key.txt (or the CENSUS_API_KEY environment variable). Then run `python ps5.py` and

`python make_report.py`.

Appendix: prompts to reproduce this analysis

#	Prompt
1	read the whole assignment pdf and help us with problem set 5 cost drivers in childcare only use the data sources the assignment points to do not hallucinate or invent any numbers on your own every number has to come from the actual data explain what you are doing in the chat as you go in simple terms and ask if anything is unclear
2	pull acs 5 year estimates for every county for 2015 through 2022 from the census api using the variable codes in the assignment table the api key is saved in a text file in the data folder
3	check that every acs variable code means the same thing in every year before using it
4	build the derived variables exactly the way the assignment defines them per capita income total population kids under 6 kids under 6 with all parents in the labor force median gross rent female college share female full time year round share under 6 share and under 6 care need share
5	from the oews zip files for 2015 to 2022 pull the mean annual wage for childcare workers occupation code 39 9011 for every metro and nonmetro area
6	zero pad the oews area codes to 7 characters and merge the wages onto counties with the county oews crosswalk on year and oews area then attach the acs data and keep only metro counties where the area code starts with 00
7	check how many counties are missing wages each year it should be around 1 percent and if it is a lot more figure out why and fix the merge
8	make a table with the mean sd and number of observations for every predictor in 2022
9	convert wages and rents to 2022 dollars using cpi u annual averages and plot how they changed over time
10	test whether real childcare worker wages were the same in 2022 as in 2019 give the hypotheses test statistic and p value and explain what it means for the providers claim
11	regress 2022 childcare worker wages on female college share report the coefficient whether we can reject that it is zero and what it means in plain words
12	put everything in a pdf with our team names winston park ayush rajpal tobin gedstad wen ding and alfred mordecai
13	now double check all the results rerun everything from the raw files and verify every number with a separate calculation and make sure the numbers in the pdf match before giving it to us